LinkedIn edunet foundation

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

[SMART HEALTH METRICS & PATIENT DATA INTELLIGENCE DASHBOARD]

Project Domain

[Data Analytics using Excel, Power BI with Copilot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

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ABSTRACT OF THE PROJECT

The project begins with collecting raw hospital patient records containing demographic details, admission and discharge dates, diagnoses, vital signs, treatment outcomes, and length of stay. Using Microsoft Excel, the dataset of over 1,500 records was systematically cleaned and organized. This process involved removing duplicate entries, correcting inconsistencies such as missing blood pressure values, handling null fields, standardizing formats, and structuring the data into meaningful tables. Proper data preparation ensured accuracy, reliability, and consistency, which are essential for effective healthcare analysis and decision-making.

After that, the cleaned dataset was analyzed using Microsoft Power BI, a business intelligence and visualization tool. Power BI was used to model relationships between patient attributes, perform calculations such as average length of stay and recovery rates, and transform raw information into actionable insights. Various charts, graphs, and interactive visuals were created to simplify complex healthcare data and make it easier to interpret.

The system focuses on tracking important Key Performance Indicators (KPIs) such as patient admissions, discharges, bed occupancy rates, length of stay, and treatment outcomes. Analysis of the dataset revealed that, diabetes and heart disease cases often involved longer hospital stays. Critical and deceased outcomes were more frequent among elderly patients, highlighting age as a significant factor in treatment results. Monitoring these metrics helps hospital administrators evaluate operational performance, identify trends, manage resources efficiently, and ensure smooth patient flow across departments.

Interactive dashboards were designed to provide dynamic insights through filters, slicers, and drill-down features. These dashboards allow users to quickly explore data, compare outcomes across diagnoses or age groups, and generate instant reports without manual calculations. This enhances transparency and enables faster, data-driven decision-making.

Overall, the solution improves hospital efficiency, reduces manual errors, saves time, and supports better planning and resource allocation. By providing accurate and timely information, the system ultimately contributes to enhanced patient care quality and improved healthcare services.

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CHAPTER 1
INTRODUCTION

Healthcare organizations generate a large volume of patient data every day, including demographic details, admission and discharge records, diagnoses, and treatment outcomes. However, this data is often unstructured, incomplete, and difficult to analyze in its raw form. Without proper preparation and analysis, hospitals struggle to make effective decisions and improve patient care.

And the solution improves hospital management by reducing manual errors, saving time, and supporting better planning and resource allocation. By transforming raw patient data into actionable insights, the system enhances transparency, enables data-driven decision-making, and ultimately contributes to improved patient care quality and healthcare services.

1.1 Problem Statement

1. Incomplete and Unstructured Data

Hospital patient records often come from multiple sources such as OPD, IPD, laboratory, pharmacy, and billing systems. In the provided dataset, issues like missing blood pressure values, inconsistent formats, and duplicate entries were observed. Such unstructured data reduces reliability and makes analysis challenging. Proper cleaning and standardization are essential to ensure accurate insights and decision-making.

2. Lack of Real-Time Visibility

The dataset reflects historical patient records, but hospitals often lack real-time dashboards to monitor admissions, discharges, bed occupancy, or treatment progress. Without instant visibility, decisions are delayed, leading to inefficiencies. Real-time monitoring would help administrators respond quickly to emergencies and manage resources effectively.

3. Manual Work and Errors

Patient data is frequently recorded manually in spreadsheets or paper files, increasing the risk of human errors such as incorrect entries or calculation mistakes. Inconsistent reporting across departments further complicates analysis. Automating data collection and analysis reduces dependency on manual processes and ensures greater precision.

4. Slow and Inconsistent Reporting

Preparing reports manually requires significant time and effort. Different departments may calculate metrics differently, resulting in inconsistent outputs. This slows down hospital management's ability to make timely decisions. Automated reporting tools like Power BI provide standardized and faster results.

5. Need for Data Cleaning and Organization

Raw healthcare data must be cleaned to remove duplicates, fill missing values, correct errors, and standardize formats. For example, several patient records in the dataset had missing systolic blood pressure values or inconsistent temperature formats. Organizing data into structured tables improves accessibility and usability, forming the foundation for reliable dashboards and insights.

6. Tracking Health Metrics

Key Performance Indicators (KPIs) such as admissions, discharges, bed occupancy rate, average length of stay, and treatment outcomes need to be tracked regularly. In the dataset, outcomes include *Recovered, Improved, Critical,* and *Deceased*. Tracking these metrics helps identify trends, gaps, and areas needing improvement.

7. Monitoring Patient Flow and Workload

Hospitals must manage patient movement from admission to discharge efficiently. The dataset shows varying lengths of stay (from 1 day to over 19 days), which directly impacts bed availability and staff workload. Monitoring patient flow ensures better utilization of beds, equipment, and medical staff.

8. Improving Efficiency and Decisions

Data-driven decision-making enables hospital management to allocate resources wisely, optimize schedules, and reduce operational costs. Analytical tools help predict future demands and identify inefficiencies. Improved efficiency ensures smoother hospital operations and better service delivery.

9. Ensuring Better Patient Care

Accurate and timely data directly impacts patient care quality. With reliable dashboards, healthcare professionals can quickly assess patient conditions, plan treatments effectively, and minimize errors. Ultimately, structured data systems enhance patient safety, satisfaction, and overall healthcare outcomes.

1.2 Motivation

1. Real-World Relevance

Hospitals generate large volumes of patient records daily, including demographics, diagnoses, vitals, and treatment outcomes. Without proper analysis, this data remains underutilized. By transforming raw records into meaningful insights, healthcare organizations can improve patient care, optimize resources, and enhance operational efficiency.

2. Personal and Professional Value

This project is meaningful because it demonstrates how modern data analytics tools like Excel, Power BI, and AI Copilot can be applied to real healthcare challenges. It provides hands-on experience in cleaning, analyzing, and visualizing complex datasets, which is valuable for both academic learning and professional growth in the field of data analytics and healthcare management.

3. Industry Alignment

Healthcare analytics is a rapidly growing domain where data-driven decision-making is becoming essential. The use of dashboards and AI-assisted reporting aligns with current industry trends, where hospitals and healthcare providers increasingly rely on technology to monitor patient flow, track

treatment outcomes, and manage resources effectively The motivation behind this project is to address real-world challenges in healthcare data management. Accurate and well-structured patient data is essential for improving diagnosis, treatment planning, and hospital management.This project aims to bridge the gap between raw data and meaningful insights by using modern tools like Power BI and AI-assisted technologies.

4. Impact on Hospital Management

By tracking Key Performance Indicators (KPIs) such as admissions, discharges, bed occupancy, length of stay, and patient outcomes, hospital administrators can identify inefficiencies, predict future demands, and make informed decisions. This directly contributes to smoother operations and better patient care.

5. Contribution to Patient Care

Reliable and timely insights help doctors and staff make faster diagnoses, plan treatments effectively, and reduce errors. The motivation behind this project is not only to improve hospital efficiency but also to ensure that patients receive safer, more effective, and higher-quality healthcare services.

1.3 Objectives

1. Collect and Understand Patient Records

Gather raw hospital data containing patient demographics, admission/discharge details, diagnoses, vital signs, and treatment outcomes. Understand the structure and scope of the dataset to identify key attributes for analysis.

2. Clean and Preprocess Data

Remove duplicate entries, correct inconsistencies such as missing blood pressure values, handle null fields, and standardize formats. Ensure the dataset is reliable, accurate, and ready for analysis.

3. Organize Data into Structured Tables

Transform raw records into meaningful categories such as admissions, diagnoses, vitals, and outcomes. Structured tables improve accessibility and form the foundation for dashboards and reports.

4. Perform Analytical Calculations

Use Excel and Power BI to calculate important metrics such as average length of stay, recovery rates, mortality rates, and disease-specific performance indicators.

5. Track Key Performance Indicators (KPIs)

Monitor hospital KPIs including patient admissions, discharges, bed occupancy rates, treatment outcomes, and workload distribution. These metrics provide insights into hospital efficiency and patient care quality.

6. Build Interactive Dashboards

Design Power BI dashboards with charts, graphs, filters, slicers, and drill-down features. Enable hospital administrators to explore data dynamically and generate instant reports.

7. Integrate AI Tools for Insights

Leverage Microsoft Copilot to generate automated summaries, highlight trends, and provide AI-assisted insights that enhance reporting and reduce manual effort.

8. Support Hospital Decision-Making

Provide actionable insights that help administrators allocate resources wisely, optimize schedules, and improve patient flow. Ensure that data-driven decisions lead to better healthcare outcomes.

9. Enhance Patient Care Quality

Ultimately, the project aims to improve patient safety, satisfaction, and treatment effectiveness by delivering accurate, timely, and transparent information through structured dashboards.

1.4 Scope of the Project

1. Dataset Coverage

The project uses hospital patient records containing demographic details, admission and discharge dates, diagnoses, vital signs (heart rate, blood pressure, blood sugar, temperature), length of stay, and treatment outcomes. The dataset includes over 1,500 entries, providing a comprehensive view of patient flow and hospital performance.

2. Tools and Technologies

The scope is limited to Microsoft Excel for data cleaning, Microsoft Power BI for visualization, and AI Copilot for automated insights and reporting. These tools are chosen for their accessibility, ease of use, and integration within the Microsoft ecosystem.

3. Focus Areas

The project emphasizes:

- Data cleaning and preprocessing (removing duplicates, correcting missing values, standardizing formats).
- KPI tracking (admissions, discharges, bed occupancy, length of stay, recovery vs. mortality rates).
- Dashboard creation with interactive filters, slicers, and drill-down features.
- AI-assisted reporting using Copilot to generate summaries and highlight trends.

4. Out of Scope

The project does not include predictive modeling, machine learning algorithms, or real-time data integration. It is designed as a fellowship showcase project rather than a production-level hospital management system.

5. Intended Use

The dashboards and insights are intended for demonstration and learning purposes. They showcase how healthcare data can be transformed into actionable insights but are not meant to replace existing hospital information systems.

6. Contribution

By defining clear boundaries, the project ensures focus on practical data cleaning, visualization, and AI-assisted reporting. The scope highlights how structured patient data can improve hospital efficiency, resource allocation, and patient care quality.

CHAPTER 2
Literature Survey

A literature survey reviews existing tools and methods used in healthcare data analysis. It helps in understanding current solutions and identifying gaps

2.1 Existing Work and Research

- **Healthcare Data Challenges**: Previous studies highlight that hospital data is often incomplete, inconsistent, and difficult to analyze. WHO guidelines emphasize the importance of standardized health data for reliable reporting.
- **Business Intelligence in Healthcare:** Research shows that dashboards built with tools like Power BI improve hospital decision-making by providing real-time visibility into patient flow, admissions, and treatment outcomes.
- **AI in Healthcare Analytics:** Studies on AI-assisted reporting demonstrate that tools like Copilot and ChatGPT can summarize patient records, detect anomalies, and reduce manual reporting time.
- **Gap Identified:** Most existing work focuses on large-scale predictive modeling, while practical, tool-based solutions for hospital administrators are less documented.

2.2 Tools and Technologies Review

- **Microsoft Excel:** Used for data cleaning, preprocessing, and structuring patient records. Functions such as conditional formatting, pivot tables, and data validation help ensure accuracy.
- **Microsoft Power BI:** Provides interactive dashboards and visualizations. It enables hospital staff to track KPIs such as admissions, discharges, length of stay, and treatment outcomes.
- **Microsoft Copilot (AI):** Assists in generating summaries, highlighting trends, and automating reporting tasks. It reduces manual effort and enhances analysis.
- **AI Chatbots (e.g., ChatGPT, Gemini):** Useful for content generation, anomaly detection, and prompt-driven analysis in healthcare reporting.

2.3 Gaps Identified

- **Limited Use of AI Copilot in Healthcare Dashboards:** While AI is widely studied in predictive analytics, its integration with Excel and Power BI for hospital performance tracking is underexplored.
- **Lack of Standardized Frameworks:** Prompt engineering best practices for healthcare reporting are still evolving and lack formal guidelines.
- **Focus on Complex Models:** Existing literature often emphasizes advanced machine learning, leaving simpler, practical solutions for non-technical hospital staff less documented.
- **Student Project Context:** Few studies demonstrate how fellowship or academic projects can apply AI-assisted dashboards to real patient datasets for learning and operational improvement.

CHAPTER 3
Proposed Methodology

This chapter explains the method used to complete the project, including system design, modules, data flow, and advantages.

3.1 System Design

The overall design of the project follows a structured flow from raw patient data to actionable insights:

- **Data Source/Input:** Patient records containing demographics, admission/discharge details, diagnoses, vitals (heart rate, blood pressure, blood sugar, temperature), length of stay, and outcomes.
- **Data Cleaning:** Using Excel to remove duplicates, handle missing values (e.g., missing systolic BP), correct inconsistencies, and standardize formats.
- **Analysis/Processing:** Applying calculations in Power BI such as average length of stay, recovery rates, and outcome distribution.
- **Visualization/Output:** Building interactive dashboards with charts, graphs, and slicers to track KPIs and provide hospital administrators with real-time insights.

3.2 Modules / Components Used

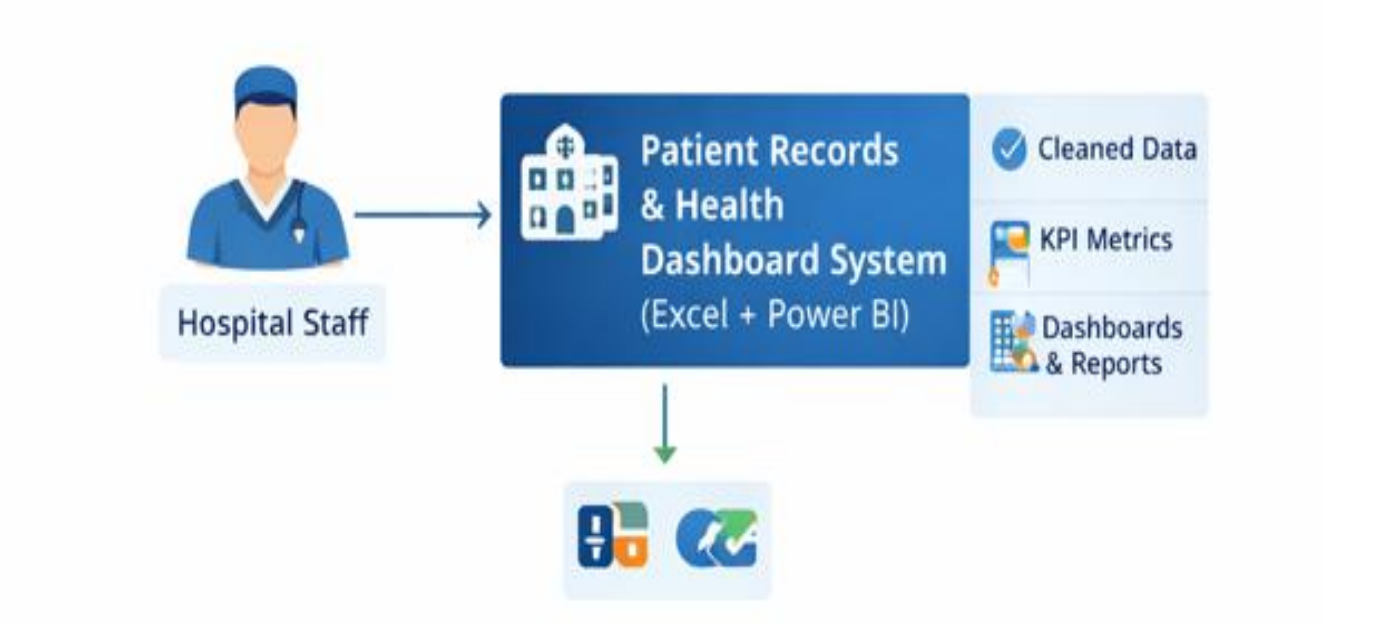
Modules:

- **Data Ingestion:** Loading raw patient records into Excel and Power BI.
  - Gathers patient data from different departments
  - Integrates multiple sources into one dataset
- **Data Cleaning:** Removing duplicates, correcting missing values, and standardizing formats.
  - Removes duplicates
  - Handles missing values
  - Corrects inconsistencies
  - Standardizes formats
- **Performing calculations:** such as admissions by diagnosis, average stay, and recovery vs. mortality rates.
  - Establishes relationships between tables
  - Creates calculated fields and measures
- **Creating dashboards**: with charts and graphs to simplify complex healthcare data.
  - Develops dashboards
  - Creates charts, graphs, slicers
  - Displays KPIs visually
- **AI Integration:** Using Copilot to generate automated summaries and highlight trends.
  - Generates automated reports
  - Provides real-time updates

3.3 Data Flow Diagram (DFD)

Raw Patient Data → Smart Health System → Dashboard & Insights.

Below is the Level 0 Data Flow Diagram:



DFD LEVEL 0 — CONTEXT

This diagram represents the overall flow of the Smart Health Metrics & Patient Data Intelligence Dashboard system. It shows how raw patient data is converted into meaningful insights.

Description:

- Raw patient dataset is taken as input from Excel
- The system performs data cleaning, analysis, and AI-based processing
- The processed data is converted into an interactive dashboard

- The final output provides business insights and reports for decision-making

Below is the Level 0 Data Flow Diagram:

DFD Level 1 – Detailed Process Flow

This diagram shows the detailed flow of data inside the Smart Health Metrics & Patient Data Intelligence Dashboard system. It breaks down the system into different processing stages.

Description:

• Raw patient dataset is first loaded and validated to ensure correct structure

• The data is then cleaned and transformed by removing duplicates, handling missing values, and correcting formats

• After cleaning, data is analyzed using Excel, Power BI, and AI tools to generate insights

• The processed data is then used to create dashboards and reports

• The final output provides clear visual insights for better understanding and decision-making

3.4 Advantages

- Automates repetitive data cleaning tasks, reducing manual effort.
- Provides clear, visual insights for non-technical hospital staff.
- Tracks KPIs such as admissions, discharges, length of stay, and outcomes.
- Leverages AI Copilot to accelerate reporting and reduce errors.
- Scalable methodology applicable to larger datasets or other healthcare domains.
- Cost-effective, using widely available Microsoft tools.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i5
RAM	Minimum 8 GB
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 11
Microsoft Excel	Used for data cleaning and preprocessing
Power BI Desktop	Used for dashboard and visualization
Microsoft Copilot	Used for AI-based analysis
AI Chatbot	Used for generating insights and report content
Web Browser	Google Chrome (latest version)

CHAPTER 4
Implementation and Results

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4.1 Data Cleaning Results

• Dataset source:
Hospital patient dataset containing demographics, admission/discharge details, diagnoses, vitals, and outcomes.

• Original dataset size:
Over 1,500 patient records.

• Issues found:

- Missing values in some fields
- Duplicate patient records
- Inconsistent formats (date, gender)
- Incorrect data entries

• Cleaning steps applied:

- Removed duplicate records
- Filled missing values where possible.
- Standardized formats (date, gender)
- Corrected data types using Excel

• Cleaned dataset size:
1500 Valid records ready for analysis.

Screenshot of the raw (uncleaned) dataset here

Patient Id	Age	Gender	Admission date	Discharge date	Diagnosis	Heart rate	Systolic bp	Diastolic bp	Blood sugar	Temperature c	Length of stay days	Outcome
1001	52	male	2024-09-11 0:00:00	2024-09-26 0:00:00	hypertension	97	92	92	151	36.3	15	Recovered
1002	15	female	2024-11-23 0:00:00	2024-11-27 0:00:00	fracture		105	66	132	35.6	4	Improved
1003	72	female	2024-02-23 0:00:00	2024-03-03 0:00:00	COVID-19	115	129	71	174	37.2	9	Deceased
1004	61	female	2024-09-05 0:00:00	2024-09-12 0:00:00	infection	110	91	75	126	35.9	7	Improved
1005	21	female	2024-08-26 0:00:00	2024-09-14 0:00:00	diabetes	57	170	75	205		19	Recovered
1006	83	female	2024-08-16 0:00:00	2024-09-04 0:00:00	diabetes	117	130	102	131	36.8	19	improved
1007	87	Male	2024-03-07 0:00:00	2024-03-13 0:00:00	heart disease	73	101	95	226	37.5	6	Improved
1008	75	male	2024-02-20 0:00:00	2024-02-26 0:00:00	infection	77	124	72	86	36.8	6	improved
1009	75	female	2024-04-06 0:00:00	2024-04-21 0:00:00	Asthma	83	154	66	135	36.8	15	recovered
1009	75	female	2024-04-06 0:00:00	2024-04-21 0:00:00	Asthma	83	154	66	135	36.8	15	recovered
1010	88	female	2024-04-01 0:00:00	2024-04-16 0:00:00	Fracture	84	144	95	75	37.5	15	recovered
1011	24	Other	2024-09-20 0:00:00	2024-09-29 0:00:00	heart disease	93	134	106	127	37.2	9	Improved
1012	3	male	2024-10-17 0:00:00	2024-10-21 0:00:00	Asthma	79	168	68	138	37.2	4	recovered
1013	22	male		2024-04-23 0:00:00	Hypertension	62	179	91	225	37.3	18	Improved
1014	53	Female	2024-03-28 0:00:00	2024-04-15 0:00:00	fracture	112	174	70	131	35.9	18	recovered
1015	2	Male	2024-03-18 0:00:00	2024-03-23 0:00:00	Hypertension	70	151	67	238	37.5	5	recovered
1016	88	female	2024-10-14 0:00:00	2024-10-25 0:00:00	infection	55	138	92	151	37.4		recovered
1016	88	female	2024-10-14 0:00:00	2024-10-25 0:00:00	infection	55	138	92	151	37.4		recovered
1017	30	female	2024-04-29 0:00:00	2024-05-04 0:00:00	covid-19	59	114	82	238	36.3	5	
1018	38	Male	2024-10-28 0:00:00	2024-11-01 0:00:00	fracture	57	112	84	246	36.8	4	Deceased
1019	2	female	2024-06-13 0:00:00	2024-06-25 0:00:00	asthma	119	113	67	246	37	12	critical
1019	2	female	2024-06-13 0:00:00	2024-06-25 0:00:00	asthma	119	113	67	246	37	12	critical
1020	64	Female	2024-06-30 0:00:00	2024-07-02 0:00:00	COVID-19	58	127	104	140	36.9	2	improved
	60	male	2024-08-10 0:00:00	2024-08-23 0:00:00	Diabetes	118	117	81	200	37.2	13	recovered
1022	21	Female	2024-11-06 0:00:00	2024-11-16 0:00:00	asthma	64	110		79	38	10	Improved
1023	33	Female	2024-01-07 0:00:00	2024-01-16 0:00:00	heart disease	75	143	99	167	37.8	9	Improved

Screenshot of the cleaned dataset or data validation summary here

Patient Id	Age	Gender	Admission date	Discharge date	Diagnosis	Heart rate	Systolic bp	Diastolic bp	Blood sugar	Temperature c	Length of stay days	Outcome
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1002	15	Female	23-11-2024	27-11-2024	Fracture	91	105	66	132	35.6	4	Improved
1003	72	Female	23-02-2024	03-03-2024	COVID-19	115	129	71	174	37.2	9	Deceased
1004	61	Female	05-09-2024	12-09-2024	Infection	110	91	75	126	35.9	7	Improved
1005	21	Female	26-08-2024	14-09-2024	Diabetes	57	170	75	205	37.1	19	Recovered
1006	83	Female	16-08-2024	04-09-2024	Diabetes	117	130	102	131	36.8	19	Improved
1007	87	Male	07-03-2024	13-03-2024	Heart Disease	73	101	95	226	37.5	6	Improved
1008	75	Male	20-02-2024	26-02-2024	Infection	77	124	72	86	36.8	6	Improved
1009	75	Female	06-04-2024	21-04-2024	Asthma	83	154	66	135	36.8	15	Recovered
1010	88	Female	01-04-2024	16-04-2024	Fracture	84	144	95	75	37.5	15	Recovered
1011	24	Other	20-09-2024	29-09-2024	Heart Disease	93	134	106	127	37.2	9	Improved
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1018	38	Male	28-10-2024	01-11-2024	Fracture	57	112	84	246	36.8	4	Deceased
1019	2	Female	13-06-2024	25-06-2024	Asthma	119	113	67	246	37	12	Critical
1020	64	Female	30-06-2024	02-07-2024	COVID-19	58	127	104	140	36.9	2	Improved
1021	60	Male	10-08-2024	23-08-2024	Diabetes	118	117	81	200	37.2	13	Recovered
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1025	58	Male	19-09-2024	26-09-2024	Diabetes	81	0	88	104	37.4	7	Recovered
1026	22	Female	09-06-2024	13-06-2024	Heart Disease	95	134	100	71	37.9	4	Improved
1027	89	Male	30-05-2024	18-06-2024	Fracture	67	162	88	193	37.3	19	Improved
1028	49	Male	25-03-2024	04-04-2024	Diabetes	63	142	93	93	37.8	10	Deceased
1029	59	Female	26-05-2024	29-05-2024	Infection	90	119	93	143	38.4	3	Recovered

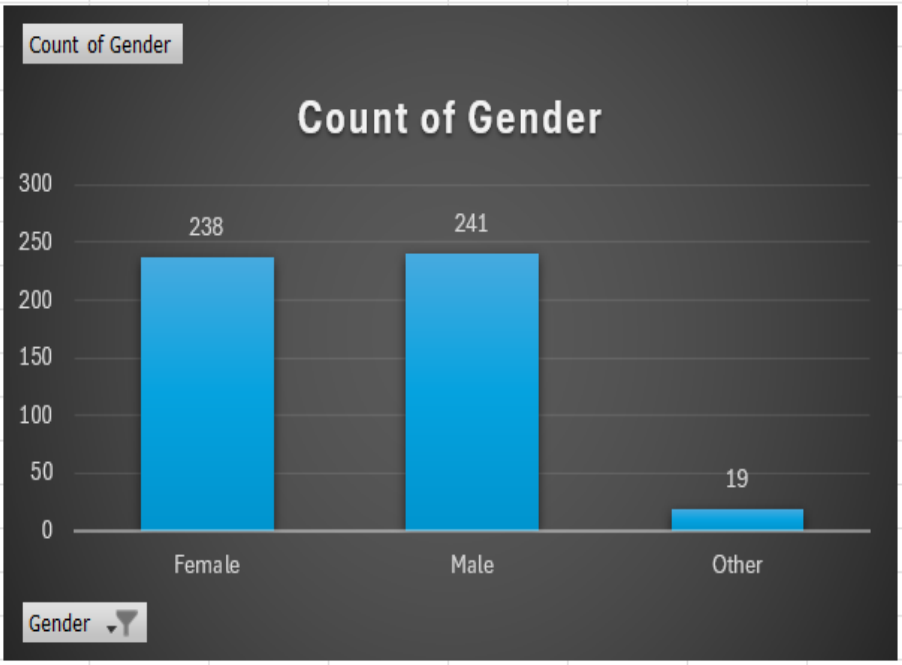
4.2 Visualization / Output Results

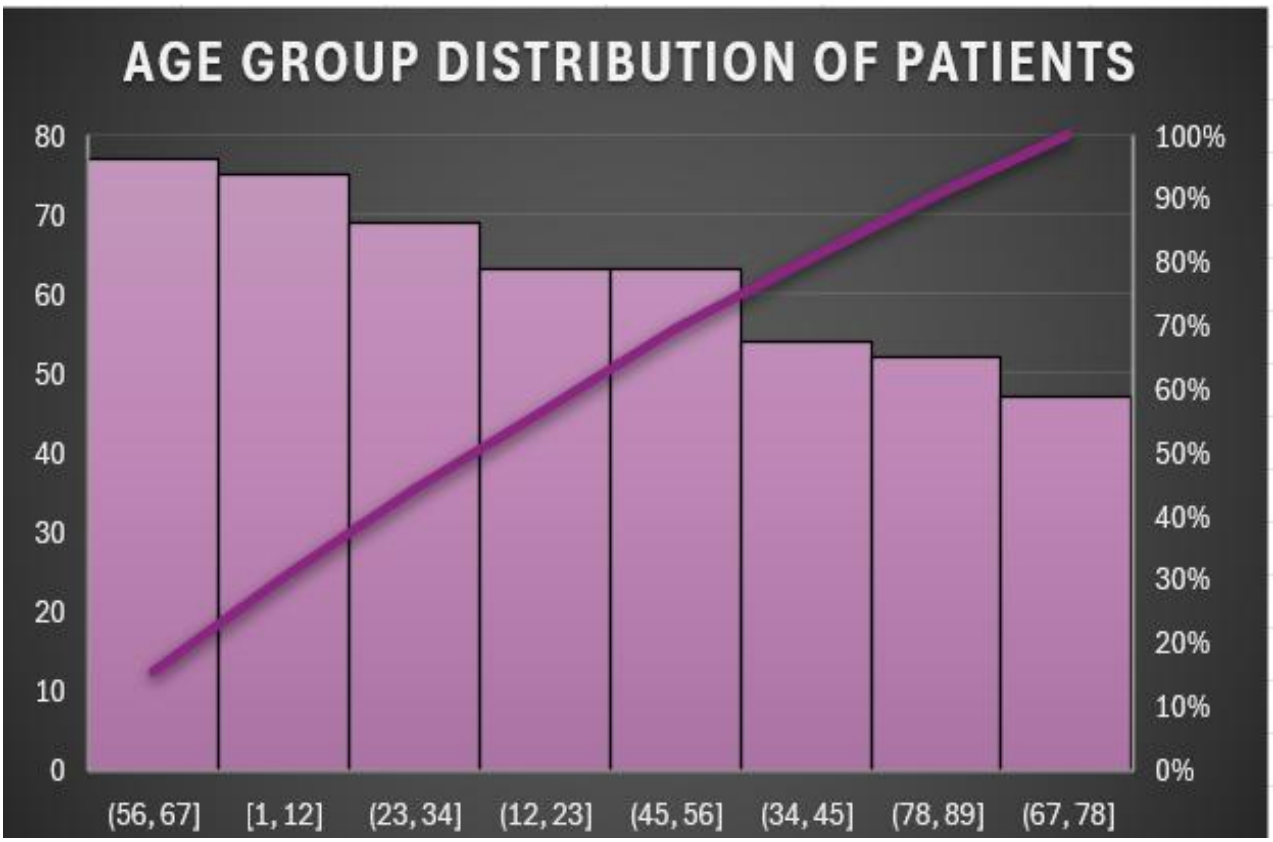
- **Dashboard / Chart 1:**
Patient demographics analysis (age and gender distribution)
- **Dashboard / Chart 2:**
Health metrics analysis (BP, sugar level, heart rate)
- **Dashboard / Chart 3:**
Admission and discharge trends over time
- **Additional Visualization:**
Disease distribution showing common health conditions

Screenshot of your main dashboard

Screenshot of additional visualization or chart here

Gender	Count of Gender
Female	238
Male	241
Other	19





4.3 AI Tool Integration Results

- **Tool used:**
Microsoft Copilot / ChatGPT
- **Prompts used:**
- "Summarize the patient dataset and identify key trends."
- "Generate insights from health metrics data."
- "Create a short report based on the dashboard."
- **Output quality:**
The AI-generated output was clear, accurate, and helpful for understanding the data.
- **How it enhanced the project:**
AI tools helped in faster analysis, reduced manual effort, and improved the quality of insights and report writing.

CHAPTER 5
Discussion and Conclusion

5.1 Key Findings

• Data cleaning improved accuracy by removing duplicate and incorrect records.

• Patient trends and health metrics helped in better understanding of hospital data.

• Dashboard provided clear insights for quick decision-making.

• AI tools reduced time and effort in analysis and report generation.

• Data visualization made it easy to identify patterns and trends.

5.2 Project Links

• GitHub Link of the Project: [Paste your GitHub link here]

• Video Recording: [Paste your video link here]

• Power BI Dashboard Link: [Paste link if available]

5.3 Limitations

• Project is based on a sample dataset

• No real-time data updates

• Limited use of advanced AI features

• Depends on tools like Excel and Power BI

5.4 Future Work

- Add real-time data integration
- Use machine learning for predictions
- Create web-based dashboard
- Expand dataset for better analysis

5.5 Conclusion

This project successfully converts hospital patient data into useful insights using Excel, Power BI, and AI tools.

The dashboard helps in better decision-making, improves data understanding, and supports hospital management. All objectives of the project were achieved, and it helped in gaining practical knowledge of data analytics and AI tools.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com

[4] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

[5] World Health Organization (WHO). Health Data and Health Information Systems. Available at: https://www.who.int/data

APPENDICES

Appendix A — Sample Dataset

Insert a sample of the raw dataset

Patient Id	Age	Gender	Admission date	Discharge date	Diagnosis	Heart rate	Systolic bp	Diastolic bp	Blood sugar	Temperature c	Length of stay days	Outcome
1001	52	male	2024-09-11 0:00:00	2024-09-26 0:00:00	hypertension	97	92	92	151	36.3	15	Recovered
1002	15	female	2024-11-23 0:00:00	2024-11-27 0:00:00	fracture		105	66	132	35.6	4	Improved
1003	72	female	2024-02-23 0:00:00	2024-03-03 0:00:00	COVID-19	115	129	71	174	37.2	9	Deceased
1004	61	female	2024-09-05 0:00:00	2024-09-12 0:00:00	infection	110	91	75	126	35.9	7	Improved
1005	21	female	2024-08-26 0:00:00	2024-09-14 0:00:00	diabetes	57	170	75	205		19	Recovered
1006	83	female	2024-08-16 0:00:00	2024-09-04 0:00:00	diabetes	117	130	102	131	36.8	19	improved
1007	87	Male	2024-03-07 0:00:00	2024-03-13 0:00:00	heart disease	73	101	95	226	37.5	6	Improved
1008	75	male	2024-02-20 0:00:00	2024-02-26 0:00:00	infection	77	124	72	86	36.8	6	improved
1009	75	female	2024-04-06 0:00:00	2024-04-21 0:00:00	Asthma	83	154	66	135	36.8	15	recovered
1009	75	female	2024-04-06 0:00:00	2024-04-21 0:00:00	Asthma	83	154	66	135	36.8	15	recovered
1010	88	female	2024-04-01 0:00:00	2024-04-16 0:00:00	Fracture	84	144	95	75	37.5	15	recovered
1011	24	Other	2024-09-20 0:00:00	2024-09-29 0:00:00	heart disease	93	134	106	127	37.2	9	Improved
1012	3	male	2024-10-17 0:00:00	2024-10-21 0:00:00	Asthma	79	168	68	138	37.2	4	recovered
1013	22	male		2024-04-23 0:00:00	Hypertension	62	179	91	225	37.3	18	Improved
1014	53	Female	2024-03-28 0:00:00	2024-04-15 0:00:00	fracture	112	174	70	131	35.9	18	recovered
1015	2	Male	2024-03-18 0:00:00	2024-03-23 0:00:00	Hypertension	70	151	67	238	37.5	5	recovered
1016	88	female	2024-10-14 0:00:00	2024-10-25 0:00:00	infection	55	138	92	151	37.4		recovered
1016	88	female	2024-10-14 0:00:00	2024-10-25 0:00:00	infection	55	138	92	151	37.4		recovered
1017	30	female	2024-04-29 0:00:00	2024-05-04 0:00:00	covid-19	59	114	82	238	36.3	5	
1018	38	Male	2024-10-28 0:00:00	2024-11-01 0:00:00	fracture	57	112	84	246	36.8	4	Deceased
1019	2	female	2024-06-13 0:00:00	2024-06-25 0:00:00	asthma	119	113	67	246	37	12	critical
1019	2	female	2024-06-13 0:00:00	2024-06-25 0:00:00	asthma	119	113	67	246	37	12	critical
1020	64	Female	2024-06-30 0:00:00	2024-07-02 0:00:00	COVID-19	58	127	104	140	36.9	2	improved
	60	male	2024-08-10 0:00:00	2024-08-23 0:00:00	Diabetes	118	117	81	200	37.2	13	recovered
1022	21	Female	2024-11-06 0:00:00	2024-11-16 0:00:00	asthma	64	110		79	38	10	Improved
1023	33	Female	2024-01-07 0:00:00	2024-01-16 0:00:00	heart disease	75	143	99	167	37.8	9	Improved
1024	76	female	2024-11-25 0:00:00	2024-12-05 0:00:00	diabetes		107	62	158	36.1	10	Recovered
1025		Male	2024-09-19 0:00:00	2024-09-26 0:00:00	diabetes	81	0	88	104	37.4	7	recovered
1026	22	female	2024-06-09 0:00:00	2024-06-13 0:00:00	heart disease	95	134	100	71	37.9	4	improved
1027	89	Male	2024-05-30 0:00:00	2024-06-18 0:00:00	fracture	67	162	88	193	37.3	19	Improved
1028	49	male	2024-03-25 0:00:00	2024-04-04 0:00:00	diabetes	63	142	93	93	37.8	10	deceased
1029	59	female	2024-05-26 0:00:00	2024-05-29 0:00:00	Infection	90	119	93	143	38.4	3	Recovered
1030	42	Male	2024-04-21 0:00:00	2024-05-10 0:00:00	Asthma	112	149	87	235	37.7	19	improved
1031	60	male		2024-02-07 0:00:00	covid-19	108	121	72	91	37.2	2	recovered
1032	80	female	2024-10-11 0:00:00	2024-10-21 0:00:00	Diabetes	66	156	74	223	37.7	10	recovered
1033	15	female	2024-07-06 0:00:00	2024-07-08 0:00:00	Asthma	68	156	82	240	37.1	2	Recovered

Sheet1

Appendix B — Key Excel Formulas / DAX Measures Used

List any important formulas, functions, or DAX measures used in the project:

- =IFERROR(VLOOKUP(...), "Not Found") — used for data validation and lookup of patient details.
- =COUNTIF(range, criteria) — used to count duplicate patient IDs or repeated entries.
- =AVERAGEIF(range, criteria, average_range) — used to calculate average vitals (e.g., average blood pressure for diabetic patients).
- =DATEDIF(AdmissionDate, DischargeDate, "d") — used to calculate length of stay in days.

- =IF(Outcome="Recovered",1,0) — used to flag recovery cases for KPI tracking.

Appendix C — AI Prompts Used

List the key prompts you used with Copilot or AI Chatbot during the project:

- Prompt 1: *"Summarize patient outcomes by diagnosis."* → Used to quickly generate insights on recovery, improvement, critical, and deceased cases across different diseases.

- Prompt 2: *"Highlight recovery vs. mortality trends across age groups."* → Helped identify how elderly patients had higher critical/deceased outcomes compared to younger groups.

- Prompt 3: *"Generate a hospital performance summary report."* → Produced a concise overview of admissions, discharges, average length of stay, and outcome distribution.

- Prompt 4: *"Identify anomalies or outliers in patient vitals and explain possible causes."* → Assisted in detecting unusual blood pressure or temperature values and linking them to potential data entry errors.

Appendix D — Screenshot Gallery

[Insert any additional screenshots, code snippets, or supplementary visuals here that were not included in the main chapters.]

Excel Formula for Length of Stay:

Code

```
=DATEDIF(AdmissionDate, DischargeDate, "d")
```

Excel Formula for Recovery Flag:

Code

```
=IF(Outcome="Recovered",1,0)
```

Power BI DAX for Recovery Rate:

Code

```
Recovery Rate =
DIVIDE(
  COUNTROWS(FILTER(Patient, Patient[Outcome] = "Recovered")),
  COUNT(Patient[PatientID])
)
```

Power BI DAX for Mortality Rate:

Code

```
Mortality Rate =
DIVIDE(
  COUNTROWS(FILTER(Patient, Patient[Outcome] = "Deceased")),
  COUNT(Patient[PatientID])
)
```